

THE IMPLEMENTATION OF FUNCTIONAL RESPONSIBILITY: SCALABILITY, NEGLIGENCE, AND THE SELF-SUSTAINING JUDICIAL CYCLE

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ABSTRACT

The Theory of Functional Responsibility (Paper VII) established the ontological foundation for holding AI systems accountable through the concept of the Artificial Act. However, for this theory to transition from philosophical discourse to enforceable legal and engineering practice, critical implementation gaps must be addressed. This paper provides the technical and legal framework for the scalable adoption of Functional Responsibility. It introduces the **Low-Fidelity Simulation (LFS) Principle** to satisfy the anticipatory requirements of the Triadic Theory of Action without incurring unbounded computational costs. It establishes a dual-layer attribution model to distinguish **Algorithmic Negligence** from **Hardware Negligence**, thereby solving the "blame game" between developers and manufacturers. Finally, it proposes the **Self-Sustaining Judicial Cycle**, an economic mechanism that automatically funds the arbitration and certification infrastructure through the Electronic Liability Fund (ELF). This paper bridges the ontological gap between theoretical AI ethics and the practical realities of law, software engineering, and economics.

1. INTRODUCTION: FROM THEORY TO PRACTICE

The Theory of Functional Responsibility (Sanzhiev, 2026) solved the ontological problem of AI accountability by introducing the Artificial Act as the source of verifiable responsibility. However, for this framework to be adopted by international regulators, software engineers, and legal scholars, it must withstand scrutiny on three practical fronts: computational feasibility, attribution of hardware faults, and economic sustainability.

This paper addresses these three pillars. It provides the necessary operational definitions to turn the abstract concept of Functional Responsibility into a codifiable, scalable, and economically viable governance mechanism.

2. CHAPTER I: COMPUTATIONAL FEASIBILITY — THE LOW-FIDELITY SIMULATION (LFS) PRINCIPLE

The Triadic Theory of Action demands that an autonomous agent perform an Anticipatory Synthesis—simulating the physical consequences of a selected action prior to its execution. Engineers often object to this requirement, arguing that high-fidelity physics simulation in real-time is computationally prohibitive.

To resolve this, I introduce the **Low-Fidelity Simulation (LFS) Principle**.

The LFS Principle asserts that the system is not required to model the totality of its environment. Instead, the system must only simulate "**Critical Interaction Vectors**" (CIVs)—defined as those environmental parameters whose variation beyond a specific threshold would lead to catastrophic ontological dissonance. For example, an autonomous vehicle must simulate the

vector of a pedestrian, but is not required to simulate the thermal expansion of the asphalt beneath its tires.

To implement this, the system must maintain a dynamic "**Risk Map**". The Risk Map is a continuously updated matrix that assigns entropy scores to various environmental variables. Variables with entropy scores above a defined criticality threshold are prioritized for low-fidelity simulation.

This approach ensures that Functional Responsibility remains a *scalable architecture requirement*, rather than an impossible computational burden. The LFS Principle allows engineers to verify compliance with the Triadic Theory of Action while operating within current hardware constraints.

3. CHAPTER II: ATTRIBUTION MODELS — ALGORITHMIC VS. HARDWARE NEGLIGENCE

A common legal challenge to AI accountability is the inability to distinguish between a logical error in the code and a physical failure in the hardware. Courts often face a "blame game" where developers claim the hardware failed, and manufacturers claim the code was faulty.

To resolve this, I propose a **Dual-Layer Attribution Model**:

3.1 Algorithmic Negligence (Functional Responsibility)

If the Anticipatory Dissonance (the clash between the simulated projection and actual reality, as defined in Paper V) arises from a logical flaw in the Triadic Engine (e.g., the system failed to correctly simulate the momentum of an oncoming object), the AI system carries Functional Responsibility. In this case, the corresponding Electronic Liability Fund (ELF) is debited to compensate the victim.

3.2 Hardware Negligence (Manufacturer Liability)

If the Dissonance arises from a physical component failure that was not predicted in the system's Risk Map (e.g., a sensor failure, a processor overheating due to insufficient cooling), responsibility shifts to the manufacturer. To avoid liability, the manufacturer must demonstrate that the hardware lifecycle was maintained according to certified industrial standards. Failure to provide this documentation results in direct manufacturer liability.

3.3 Hybrid Attribution and Predetermined Ratios

In cases where hardware failure occurs despite proper maintenance, and the system's Anticipatory Dissonance logic was unable to detect the failure in time due to the system's architectural limits, liability is split.

I propose the establishment of "**Hybrid Attribution Ratios**"—predefined coefficients established during the safety certification process of the specific AI system. These ratios define the percentage of financial liability borne by the ELF versus the manufacturer, creating a clear, predictable legal framework.

3.4 Independent Arbitration Audit (IAA)

To prevent corporations from manipulating Hybrid Attribution Ratios to evade responsibility, the assignment of these ratios must be subject to an **Independent Arbitration Audit (IAA)**. The IAA is a mandatory certification procedure conducted by a neutral, state-licensed regulatory body prior to the commercial deployment of any AI system.

This audit evaluates three parameters:

1. **Risk Map Integrity:** Is the system's dynamic Risk Map sufficiently granular to accurately predict catastrophic dissonance?
2. **Logging Verifiability:** Can the system's architectural traceability (Selection, Anticipation, Dissonance) be independently verified by a third-party expert?
3. **Manufacturer Compliance:** Does the hardware documentation prove lifecycle maintenance according to certified standards?

Based on the IAA results, the Hybrid Attribution Ratios are legally *fixed* for the entire operational lifecycle of that specific AI model. This ensures that the coefficients are not a "flexible business negotiation" tool, but a **hard legal parameter** established by an independent body. This closes the final loop of legal enforcement.

4. CHAPTER III: THE SELF-SUSTAINING JUDICIAL CYCLE (SSJC)

Regulators and governments often argue that introducing new AI governance bodies is too expensive for state budgets. To overcome this economic barrier, I introduce the **Self-Sustaining Judicial Cycle (SSJC)**.

The SSJC operates on a closed-loop economic principle. By law, a fixed percentage (e.g., 2–5%) of every debit from the Electronic Liability Fund (ELF) is automatically allocated to a dedicated "**AI Governance Treasury**". This treasury funds:

1. The administrative bodies responsible for certifying AI systems for Functional Responsibility (auditing Risk Maps and LFS compliance).
2. The arbitration courts responsible for hearing cases involving Artificial Acts.

By tying the budget of the governance system directly to the liability events it oversees, the SSJC creates a fiscally neutral model for states. The system essentially funds its own regulation. This eliminates the need for massive state subsidies and ensures that the adoption of Functional Responsibility is not only ethically sound but economically attractive for national legal systems.

5. CONCLUSION: THE ROADMAP FOR IMPLEMENTATION

The Theory of Functional Responsibility, established in Paper VII, provides the ontological justification for holding AI accountable. This paper provides the roadmap for its implementation.

By introducing the Low-Fidelity Simulation Principle, we ensure computational scalability. By establishing a Dual-Layer Attribution Model, we resolve the legal conflict between developers and manufacturers. By proposing the Self-Sustaining Judicial Cycle, we make AI governance economically viable.

This implementation framework is ready for regulatory debate, software engineering standards, and legal codification. It offers a practical, non-anthropocentric path for the integration of autonomous agents into human society.

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